

Trabalho de Macroeconometria - Trabalho 2

Pass-through cambial e política monetária no Brasil: uma análise VAR/VECM

Alunos: Guilherme Santos de Carvalho e Lucas Fortuna Arantes Junqueira Professora: Susan Schommer

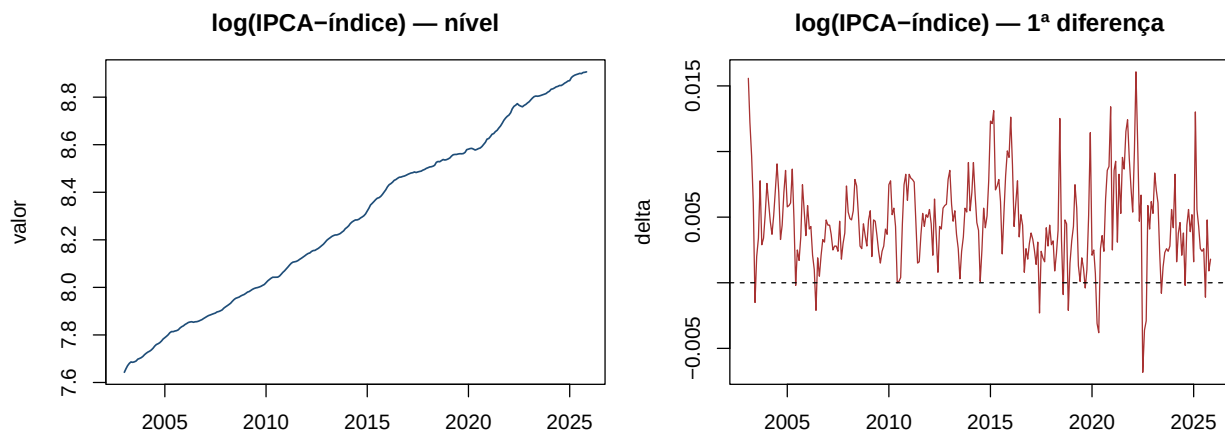
Maio de 2026

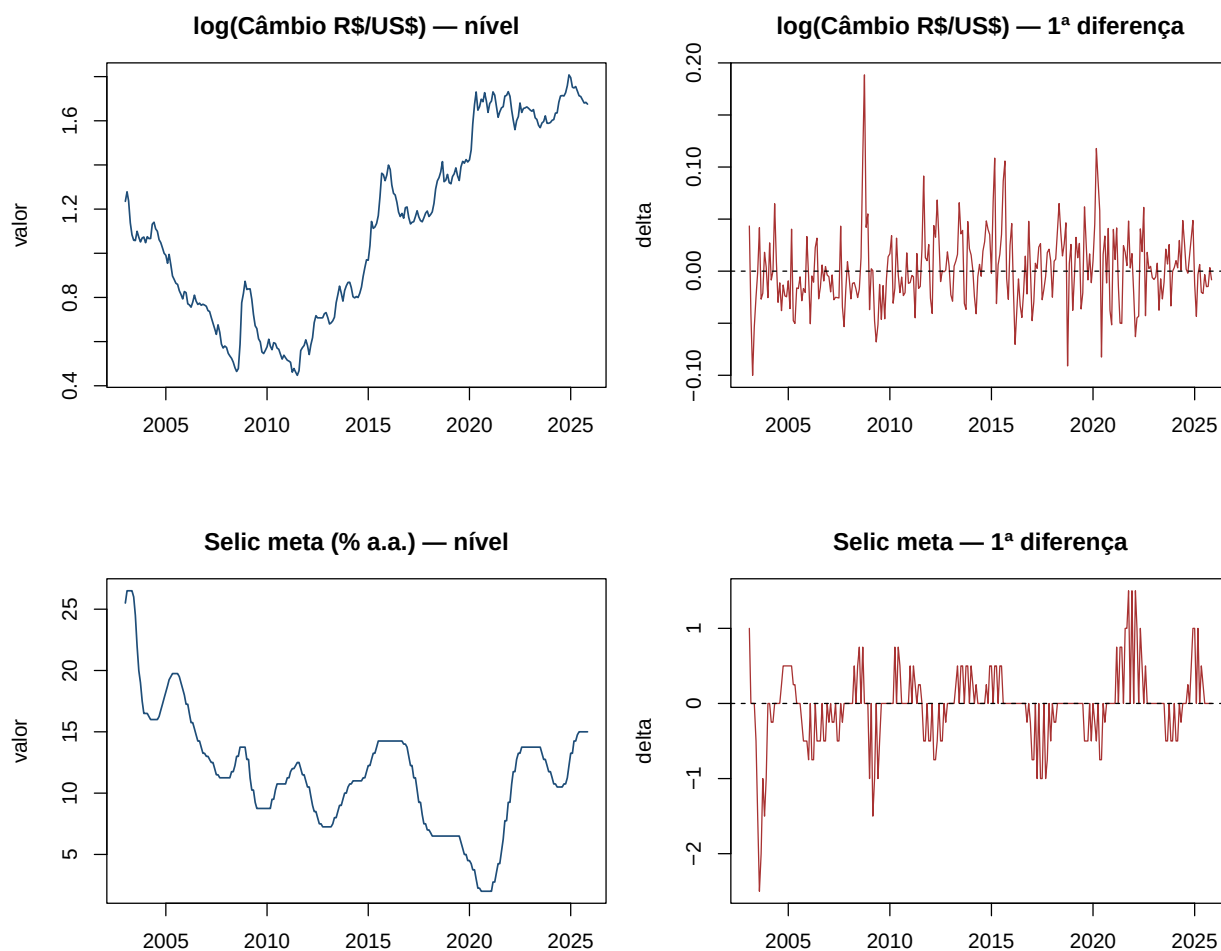
1. Estatísticas Descritivas

Table 1: Estatísticas descritivas das séries transformadas

variavel	n	media	dp	min	max
log_ipca_indice	275	8.2893	0.3696	7.6429	8.9064
log_cambio	275	1.1029	0.4122	0.4472	1.8078
selic	275	11.6018	4.6567	2.0000	26.5000

2. Visualização das Séries (Item 1)





3. Testes de Raiz Unitária (Item 2)

3.1 Teste ADF — Nível e Primeira Diferença

Table 2: Teste ADF — séries em nível e 1ª diferença

Serie	Tipo	Estatística	CV 1%	CV 5%	CV 10%	Rejeita H0 (5%)
log(IPCA) — nível	trend	-2.3018	-3.98	-3.42	-3.13	não
log(Câmbio) — nível	trend	-3.0572	-3.98	-3.42	-3.13	não
Selic — nível	drift	-3.5339	-3.44	-2.87	-2.57	SIM
log(IPCA) — 1ª dif.	drift	-7.7929	-3.44	-2.87	-2.57	SIM
log(Câmbio) — 1ª dif.	drift	-10.2321	-3.44	-2.87	-2.57	SIM
Selic — 1ª dif.	drift	-5.2617	-3.44	-2.87	-2.57	SIM

3.2 Teste Zivot-Andrews (Quebra Estrutural Endógena)

Table 3: Teste Zivot-Andrews (quebra endógena) — séries em nível

Serie	Estatística	CV 5%	Quebra	Rejeita H0 (5%)
log(IPCA) — nível	-2.4918	-5.08	2014-11-01	não
log(Câmbio) — nível	-3.2783	-5.08	2004-06-01	não
Selic — nível	-3.4451	-5.08	2017-05-01	não

Table 4: Teste Zivot-Andrews — séries em 1ª diferença

Serie	Estatística	CV 5%	Quebra	Rejeita H0 (5%)
log(IPCA) — 1ª dif.	-9.7123	-5.08	2020-08-01	SIM
log(Câmbio) — 1ª dif.	-12.2386	-5.08	2015-09-01	SIM
Selic — 1ª dif.	-10.5773	-5.08	2003-09-01	SIM

4. Testes de Cointegração (Item 3)

4.1 Engle-Granger

Table 5: Teste de Engle-Granger (ADF nos resíduos da regressão de cointegração)

Variável dependente	Estatística	CV 5%	Rejeita H0 (5%)
log(IPCA)	-1.9656	-1.95	SIM
log(Câmbio)	-2.5300	-1.95	SIM
Selic	-2.2389	-1.95	SIM

4.2 Johansen (Traço)

```
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 0.17587300888189053016397 0.06174716080533190071256
## [3] 0.01981374280892196984571 0.00000000000000000784241
##
## Values of teststatistic and critical values of test:
##
##          test 10pct  5pct  1pct
## r <= 2 |   5.40   7.52   9.24 12.97
## r <= 1 |  22.61  17.85  19.96 24.60
## r = 0  |  74.84  32.00  34.91 41.07
```

```
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          log_ipca_indice.l5 log_cambio.l5      selic.l5      constant
## log_ipca_indice.l5      1.00000000      1.00000000  1.00000000  1.00000000
## log_cambio.l5          -0.50357305     -0.3603315  -0.77399583  0.83427548
## selic.l5                -0.03151704      0.3837236   0.02244266  0.03882963
## constant                -9.48663991    -12.8087895  -7.69786528 -9.53695282
##
## Weights W:
## (This is the loading matrix)
##
##          log_ipca_indice.l5 log_cambio.l5      selic.l5
## log_ipca_indice.d      -0.0012569375 -0.0001812406 -0.0006398879
## log_cambio.d           -0.0005502148 -0.0021040137  0.0244690757
## selic.d                 0.0745800716 -0.0493376355 -0.0895917438
##
##          constant
## log_ipca_indice.d  0.00000000000000003207369
## log_cambio.d      -0.000000000000000039560366
## selic.d            0.000000000000000108225476
```

5. VAR — Seleção de Ordem e Estimação (Item 4)

Table 6: Critérios de seleção de ordem do VAR (lag.max = 12)

Critério	Defasagem ótima
AIC(n)	4
HQ(n)	3
SC(n)	3
FPE(n)	4

```
##
## VAR Estimation Results:
## =====
## Endogenous variables: log_cambio, log_ipca_indice, selic
## Deterministic variables: const
## Sample size: 270
## Log Likelihood: 1666.84
## Roots of the characteristic polynomial:
## 0.8202 0.7444 0.7444 0.685 0.685 0.5921 0.5921 0.5511 0.5511 0.509 0.509 0.4512
## Call:
## VAR(y = dY_econ, p = p, type = "const")
##
##
## Estimation results for equation log_cambio:
## =====
## log_cambio = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2 + selic.l2
```

```

##
##               Estimate Std. Error t value      Pr(>|t|)
## log_cambio.l1    0.3555429  0.0625327   5.686 0.0000000353 ***
## log_ipca_indice.l1 0.1205613  0.7724431   0.156   0.876
## selic.l1         0.0029835  0.0059043   0.505   0.614
## log_cambio.l2   -0.0928395  0.0663905  -1.398   0.163
## log_ipca_indice.l2 -0.5646268  0.8664321  -0.652   0.515
## selic.l2        -0.0018382  0.0057868  -0.318   0.751
## log_cambio.l3    0.0445689  0.0670877   0.664   0.507
## log_ipca_indice.l3 0.8170570  0.8713030   0.938   0.349
## selic.l3         0.0053109  0.0056788   0.935   0.351
## log_cambio.l4   -0.0474660  0.0624404  -0.760   0.448
## log_ipca_indice.l4 -1.0414234  0.7797922  -1.336   0.183
## selic.l4        -0.0006486  0.0056134  -0.116   0.908
## const           0.0050157  0.0050957   0.984   0.326
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.03385 on 257 degrees of freedom
## Multiple R-Squared: 0.1259, Adjusted R-squared: 0.0851
## F-statistic: 3.085 on 12 and 257 DF, p-value: 0.0004274
##
##
## Estimation results for equation log_ipca_indice:
## =====
## log_ipca_indice = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2
##
##               Estimate Std. Error t value      Pr(>|t|)
## log_cambio.l1    0.00204434  0.00501884   0.407   0.6841
## log_ipca_indice.l1 0.48626220  0.06199575   7.843 0.0000000000000118 ***
## selic.l1         0.00068476  0.00047387   1.445   0.1497
## log_cambio.l2    0.00147817  0.00532846   0.277   0.7817
## log_ipca_indice.l2 0.05074309  0.06953925   0.730   0.4662
## selic.l2         0.00048946  0.00046445   1.054   0.2929
## log_cambio.l3   -0.00073226  0.00538441  -0.136   0.8919
## log_ipca_indice.l3 -0.02469635  0.06993018  -0.353   0.7243
## selic.l3         0.00008526  0.00045578   0.187   0.8518
## log_cambio.l4    0.01248545  0.00501142   2.491   0.0134 *
## log_ipca_indice.l4 -0.07660262  0.06258559  -1.224   0.2221
## selic.l4        -0.00043543  0.00045053  -0.966   0.3347
## const           0.00256133  0.00040897   6.263 0.000000001579909 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.002717 on 257 degrees of freedom
## Multiple R-Squared: 0.3319, Adjusted R-squared: 0.3007
## F-statistic: 10.64 on 12 and 257 DF, p-value: < 0.00000000000000022
##
##
## Estimation results for equation selic:
## =====
## selic = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2 + selic.l1

```

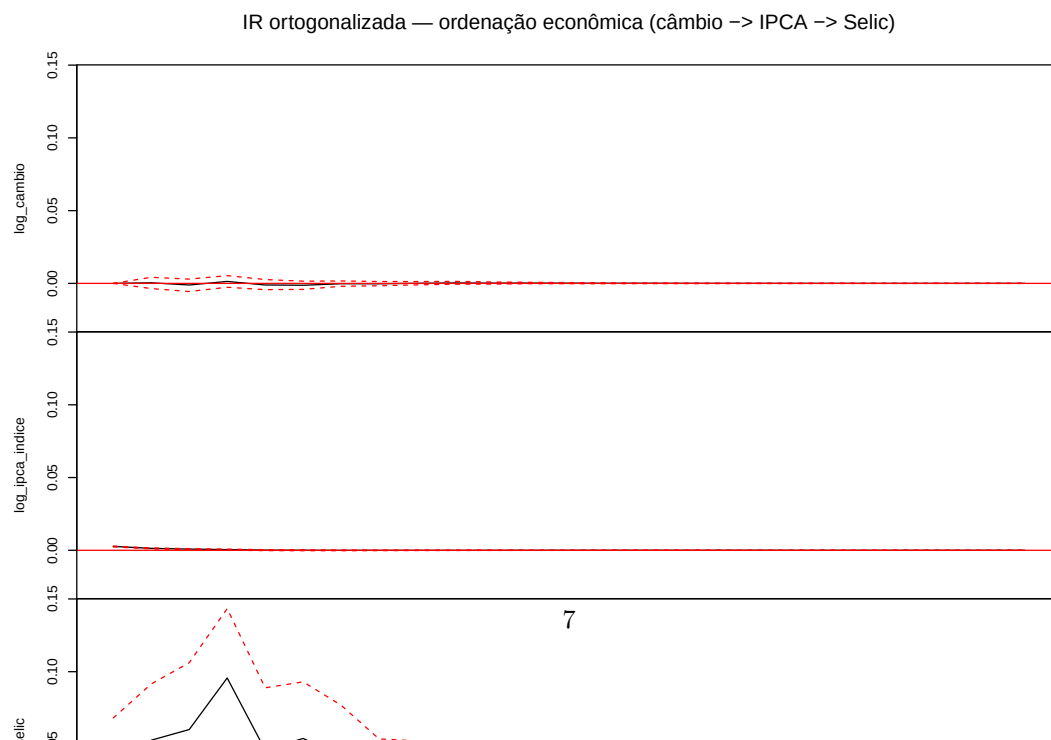
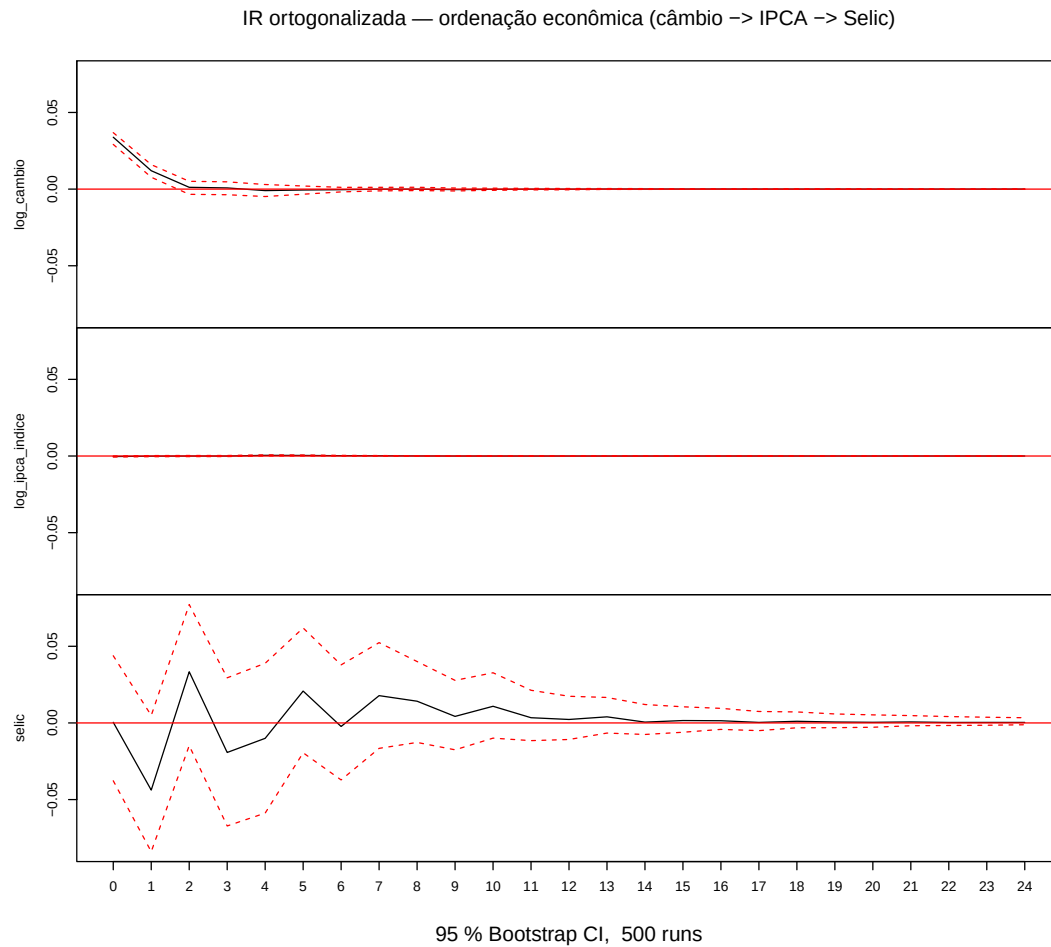
```

##
##               Estimate Std. Error t value    Pr(>|t|)
## log_cambio.l1    -1.12529    0.64519  -1.744    0.08233 .
## log_ipca_indice.l1 17.11969    7.96977   2.148    0.03264 *
## selic.l1         0.26935    0.06092   4.422 0.000014475 ***
## log_cambio.l2     1.83639    0.68499   2.681    0.00782 **
## log_ipca_indice.l2  5.76080    8.93951   0.644    0.51988
## selic.l2         0.32565    0.05971   5.454 0.000000115 ***
## log_cambio.l3    -0.88879    0.69218  -1.284    0.20029
## log_ipca_indice.l3 11.36701    8.98977   1.264    0.20722
## selic.l3         0.30661    0.05859   5.233 0.000000346 ***
## log_cambio.l4     0.12135    0.64424   0.188    0.85074
## log_ipca_indice.l4 -12.93200    8.04559  -1.607    0.10921
## selic.l4        -0.18483    0.05792  -3.191    0.00159 **
## const           -0.10690    0.05257  -2.033    0.04305 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.3493 on 257 degrees of freedom
## Multiple R-Squared:  0.5177, Adjusted R-squared:  0.4952
## F-statistic: 22.99 on 12 and 257 DF, p-value: < 0.00000000000000022
##
##
## Covariance matrix of residuals:
##               log_cambio log_ipca_indice    selic
## log_cambio      0.00114606   -0.000011524 0.00001413
## log_ipca_indice -0.00001152    0.000007382 0.00006757
## selic           0.00001413    0.000067573 0.12200200
##
## Correlation matrix of residuals:
##               log_cambio log_ipca_indice    selic
## log_cambio      1.000000    -0.1253 0.001195
## log_ipca_indice -0.125285      1.0000 0.071202
## selic           0.001195      0.0712 1.000000

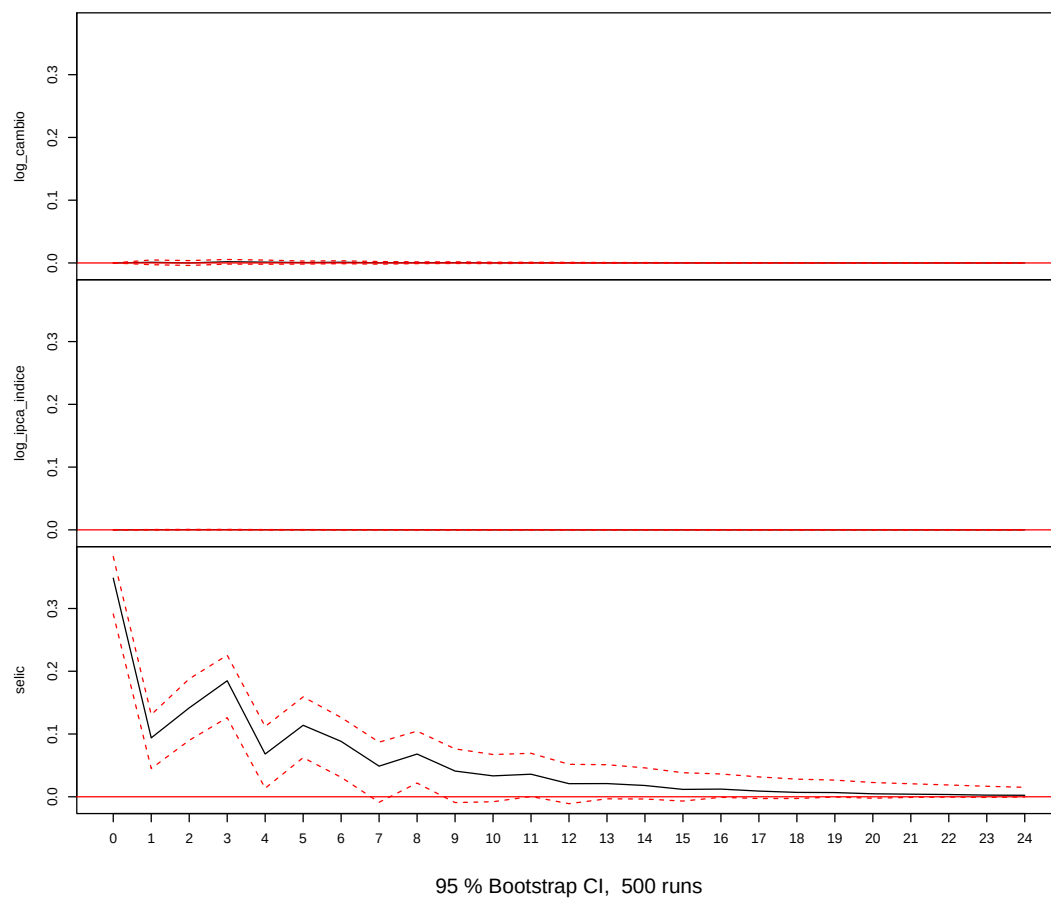
```

6. Funções de Impulso-Resposta (Item 5)

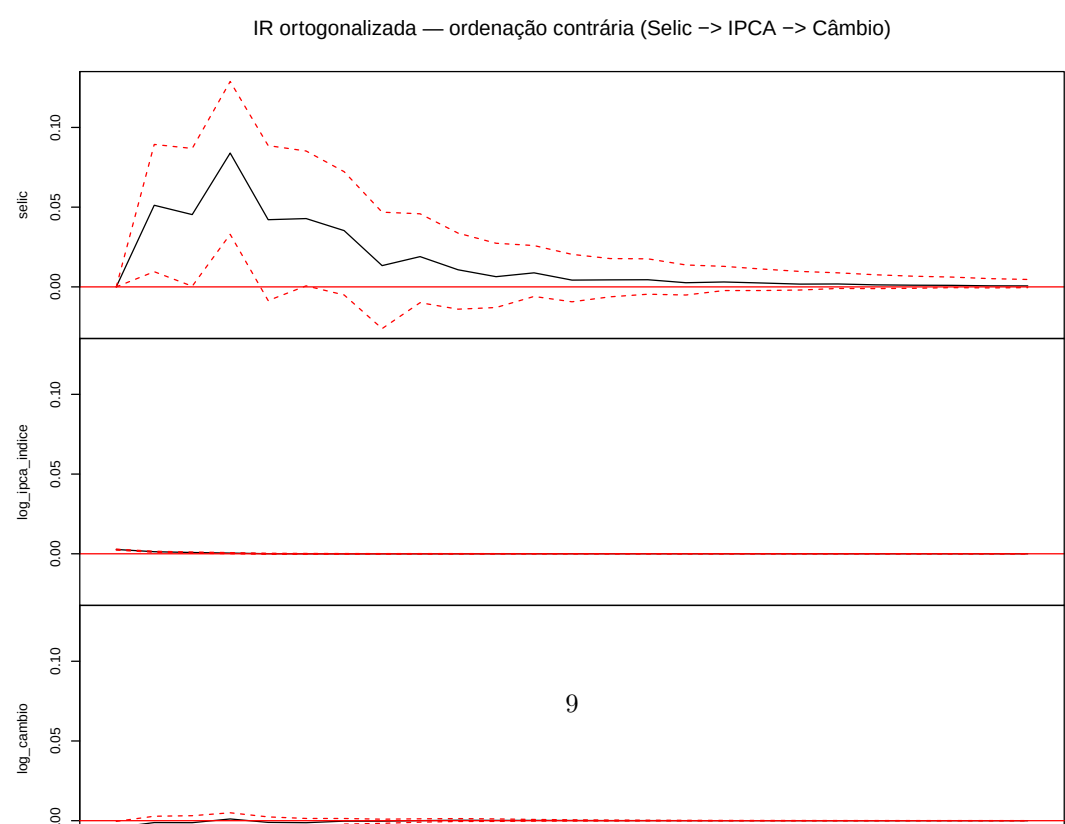
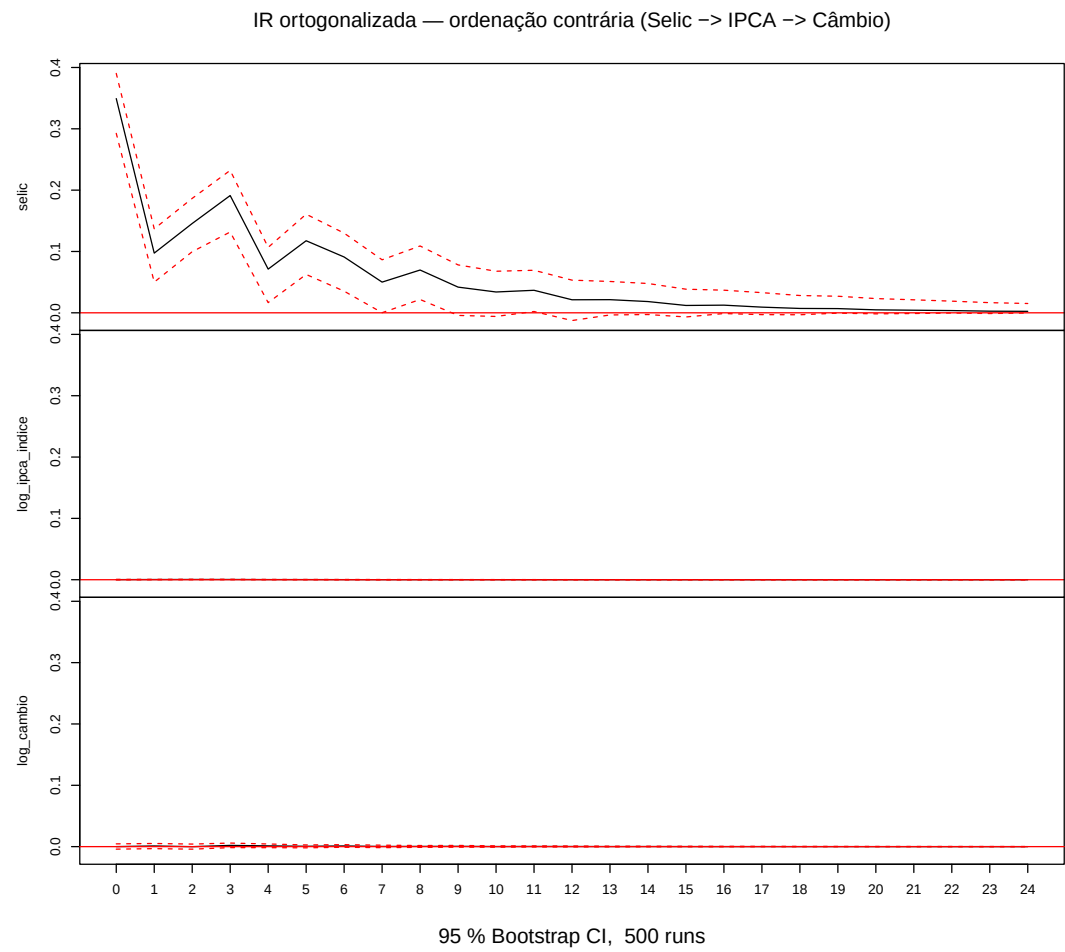
6.1 Ordenação Econômica: Câmbio → IPCA → Selic

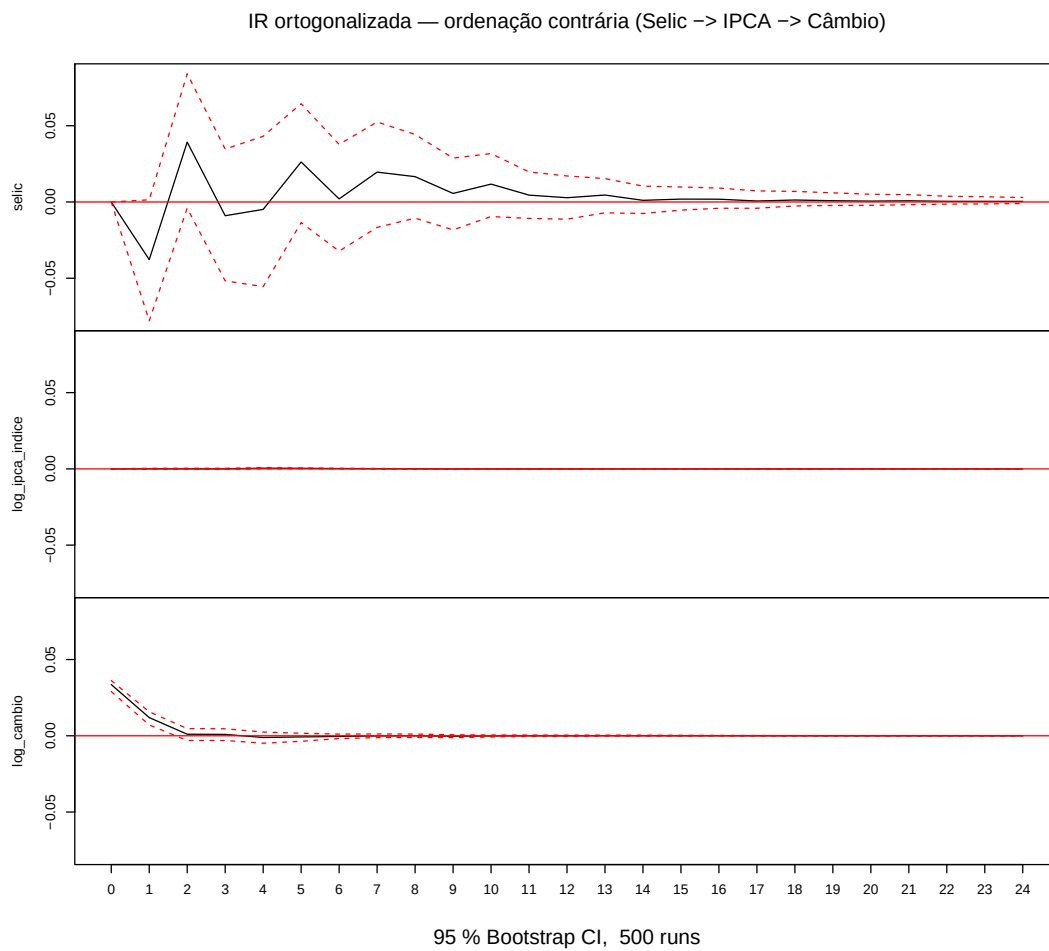


IR ortogonalizada — ordenação econômica (câmbio -> IPCA -> Selic)

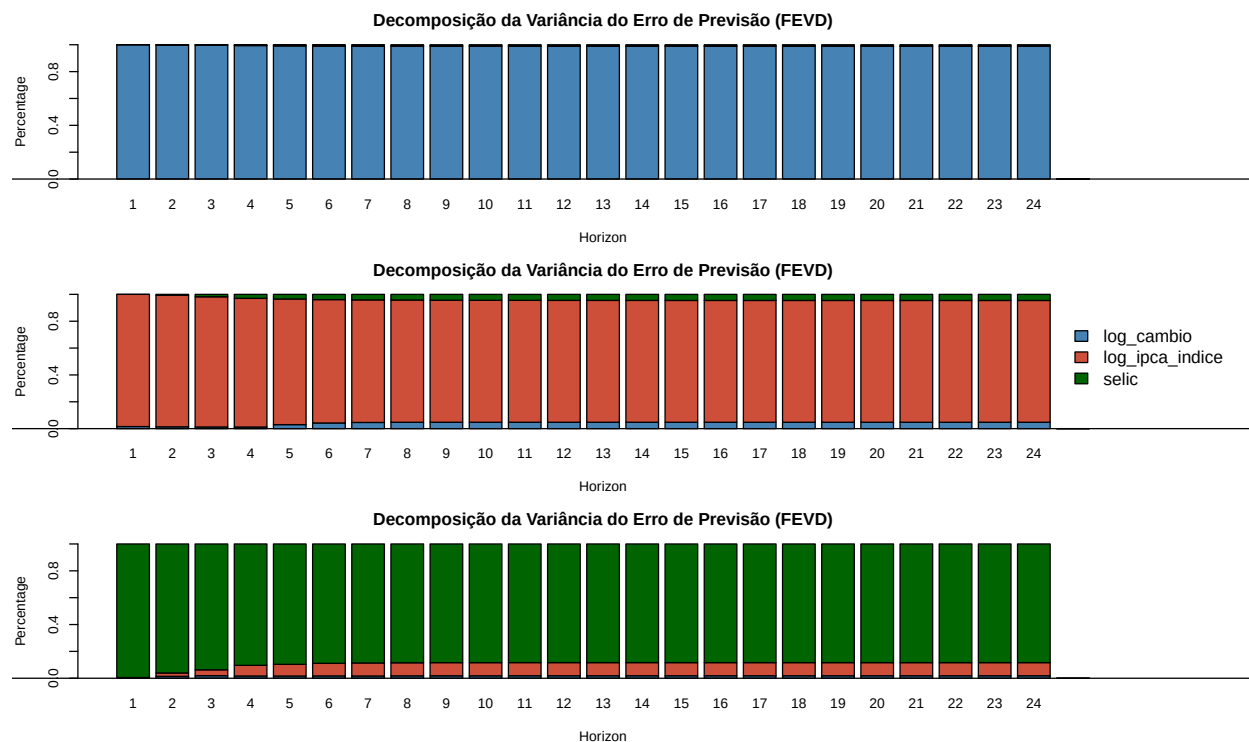


6.2 Ordenação Contrária: Selic → IPCA → Câmbio (Robustez)





6.3 Decomposição da Variância do Erro de Previsão (FEVD)



7. VECM — Estimação (Item 6)

7.1 Rank de Cointegração Adotado

Rank de cointegração adotado: 2

7.2 Vetor de Cointegração ()

Table 7: Vetor de cointegração normalizado ()

	ect1	ect2
log_ipca_indice.l5	1.0000	0.0000
log_cambio.l5	0.0000	1.0000
selic.l5	1.4283	2.8989
constant	-21.1658	-23.1926

7.3 Coeficientes de Ajustamento ()

Table 8: Coeficientes de ajustamento ()

	log_ipca_indice.d	log_cambio.d	selic.d
ect1	-0.0014	-0.0027	0.0252
ect2	0.0007	0.0010	-0.0198

7.4 Sumário das Equações do VECM

```
## Response log_ipca_indice.d :
##
## Call:
## lm(formula = log_ipca_indice.d ~ ect1 + ect2 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0109874 -0.0014052 -0.0001703  0.0015524  0.0086781
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## ect1          -0.00143818  0.00022614  -6.360 0.000000000924013 ***
## ect2           0.00069827  0.00010668   6.545 0.000000000322348 ***
## log_ipca_indice.dl1  0.47274456  0.06200814   7.624 0.0000000000000481 ***
## log_cambio.dl1      0.00274739  0.00500489   0.549      0.58352
## selic.dl1          0.00070588  0.00049321   1.431      0.15360
## log_ipca_indice.dl2  0.04469146  0.06918709   0.646      0.51889
## log_cambio.dl2      0.00211564  0.00532281   0.397      0.69135
## selic.dl2          0.00053549  0.00046982   1.140      0.25545
## log_ipca_indice.dl3 -0.03146291  0.06960838  -0.452      0.65165
## log_cambio.dl3     -0.00002692  0.00535743  -0.005      0.99600
## selic.dl3          0.00016672  0.00045499   0.366      0.71435
## log_ipca_indice.dl4 -0.09023507  0.06315728  -1.429      0.15430
## log_cambio.dl4      0.01330407  0.00501069   2.655      0.00842 **
## selic.dl4         -0.00036271  0.00045152  -0.803      0.42254
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002701 on 256 degrees of freedom
## Multiple R-squared:  0.7764, Adjusted R-squared:  0.7642
## F-statistic: 63.5 on 14 and 256 DF,  p-value: < 0.00000000000000022
##
##
## Response log_cambio.d :
##
## Call:
## lm(formula = log_cambio.d ~ ect1 + ect2 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
```

```

## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.103702 -0.020425 -0.001523  0.018308  0.146459
##
## Coefficients:
##              Estimate Std. Error t value    Pr(>|t|)
## ect1            -0.00265423  0.00283302  -0.937      0.350
## ect2             0.00103522  0.00133650   0.775      0.439
## log_ipca_indice.dl1  0.20121429  0.77681727   0.259      0.796
## log_cambio.dl1      0.34992235  0.06269964  5.581 0.0000000609 ***
## selic.dl1         -0.00004754  0.00617883  -0.008      0.994
## log_ipca_indice.dl2 -0.49430227  0.86675281  -0.570      0.569
## log_cambio.dl2     -0.10069222  0.06668241  -1.510      0.132
## selic.dl2         -0.00362532  0.00588575  -0.616      0.538
## log_ipca_indice.dl3  0.89621039  0.87203066   1.028      0.305
## log_cambio.dl3      0.04108806  0.06711612   0.612      0.541
## selic.dl3          0.00468036  0.00569996   0.821      0.412
## log_ipca_indice.dl4 -0.76750605  0.79121336  -0.970      0.333
## log_cambio.dl4     -0.05549552  0.06277222  -0.884      0.377
## selic.dl4          0.00033274  0.00565648   0.059      0.953
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03384 on 256 degrees of freedom
## Multiple R-squared:  0.1336, Adjusted R-squared:  0.08622
## F-statistic:  2.82 on 14 and 256 DF,  p-value: 0.000607
##
##
## Response selic.d :
##
## Call:
## lm(formula = selic.d ~ ect1 + ect2 + log_ipca_indice.dl1 + log_cambio.dl1 +
##      selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 + selic.dl2 +
##      log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 + log_ipca_indice.dl4 +
##      log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.66009 -0.20066 -0.00764  0.18245  1.14891
##
## Coefficients:
##              Estimate Std. Error t value    Pr(>|t|)
## ect1             0.02524    0.02838   0.889    0.37464
## ect2            -0.01978    0.01339  -1.477    0.14086
## log_ipca_indice.dl1 16.46214    7.78248   2.115    0.03537 *
## log_cambio.dl1     -1.41603    0.62815  -2.254    0.02502 *
## selic.dl1          0.19386    0.06190   3.132    0.00194 **
## log_ipca_indice.dl2  6.21650    8.68349   0.716    0.47471
## log_cambio.dl2      1.52200    0.66805   2.278    0.02354 *
## selic.dl2          0.28130    0.05897   4.771 0.000003088 ***
## log_ipca_indice.dl3 12.27961    8.73636   1.406    0.16106
## log_cambio.dl3     -1.10193    0.67240  -1.639    0.10248
## selic.dl3          0.29015    0.05710   5.081 0.000000724 ***
## log_ipca_indice.dl4 -8.21482    7.92670  -1.036    0.30102

```

```
## log_cambio.dl4      -0.19954    0.62888  -0.317    0.75128
## selic.dl4          -0.15939    0.05667  -2.813    0.00529 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.339 on 256 degrees of freedom
## Multiple R-squared:  0.5509, Adjusted R-squared:  0.5264
## F-statistic: 22.43 on 14 and 256 DF,  p-value: < 0.00000000000000022
```

8. Diagnósticos de Especificação (Item 7)

8.1 Diagnósticos do VAR

Table 9: Diagnósticos do VAR

Teste	Estatística	p-valor	H0
Portmanteau (lags=12)	87.5455	0.1026	Resíduos são ruído branco
Jarque-Bera multivariado	NA	NA	Resíduos normais
Breusch-Pagan — log(câmbio)	54.6245	0.0000	Homocedasticidade
Breusch-Pagan — log(IPCA)	0.4185	0.8112	Homocedasticidade
Breusch-Pagan — Selic	5.6524	0.0592	Homocedasticidade
Estabilidade	0.8202	NA	Estável se $\max raiz < 1$

8.2 Diagnósticos do VECM

Table 10: Diagnósticos do VECM

Teste	Estatística	p-valor	H0
Portmanteau (lags=12)	89.2013	0.0302	Resíduos são ruído branco
Jarque-Bera multivariado	NA	NA	Resíduos normais
Breusch-Pagan — log(câmbio)	8.6248	0.0134	Homocedasticidade
Breusch-Pagan — log(IPCA)	1.4859	0.4757	Homocedasticidade
Breusch-Pagan — Selic	30.4372	0.0000	Homocedasticidade
Estabilidade	1.0000	NA	Estável se $\max raiz < 1$